

Emotional Stress and Its Predictors Among UiTM Kota Bharu Students

Statistical analysis project | Bachelor of Science (Hons.) Statistics | UiTM Kota Bharu

1. Project Overview

This study examined emotional stress among full-time undergraduate students at Universiti Teknologi MARA (UiTM) Kota Bharu and investigated unemployment anxiety, academic pressure and financial strain as potential factors associated with emotional stress.

2. Research Objectives

- Determine significant differences in emotional stress among students across semesters.
- Assess the relationships between unemployment anxiety, academic pressure, financial strain, and emotional stress.
- Examine the significant effects of unemployment anxiety, academic pressure, and financial strain on emotional stress.

3. Data and Study Design

Study design	Quantitative, cross-sectional survey
Target population	Full-time undergraduate students at UiTM Kota Bharu
Main sample	298 respondents
Data collection	Structured self-administered questionnaire
Measurement	5-point Likert scale for core study variables

4. Tools and Statistical Methods

Tools: SPSS and Microsoft Excel

Methods: descriptive statistics, reliability analysis using Cronbach's alpha, one-way ANOVA, Pearson correlation, and stepwise multiple linear regression.

5. Reliability Results

The questionnaire demonstrated strong internal consistency in the main study. All constructs exceeded the commonly used 0.70 threshold.

Construct	Cronbach's α
Emotional Stress	0.944
Unemployment Anxiety	0.870
Academic Pressure	0.821
Financial Strain	0.901

6. Key Statistical Results

Pearson correlation: Emotional stress was positively related to unemployment anxiety ($r = 0.505$), academic pressure ($r = 0.702$), and financial strain ($r = 0.636$), with all relationships significant at the 0.01 level.

One-way ANOVA: Emotional stress differed significantly overall across semesters ($p = 0.034$). Semester 2 had the lowest means of emotional stress (2.6000), while Semester 7 had the highest (3.4143). However, Tukey HSD found no significant pairwise differences.

7. Stepwise Regression — Main Finding

The final stepwise regression model retained **academic pressure** and **financial strain** as significant predictors of emotional stress. Unemployment anxiety was excluded from the final model because its p -value increased to 0.058.

Final model: **$R = 0.762$** , **$R^2 = 0.581$** , and **Adjusted $R^2 = 0.578$** . Thus, the final model explained **57.8% of the variance** in emotional stress after adjustment.

Predictor	Standardized β	p-value
Academic Pressure	0.505	< 0.001
Financial Strain	0.356	< 0.001

8. Key Takeaway

Among the factors examined, academic pressure showed the strongest contribution to emotional stress in the final regression model, followed by financial strain. The findings highlight the importance of academic and financial support when addressing emotional stress among university students.

9. Data Privacy

The original survey dataset is not included in this portfolio to protect the privacy and confidentiality of respondents.

Source: Final Year Project report, UiTM Kota Bharu (2026).